VRAJ VAGHELA

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PROFESSIONAL SUMMARY

Final-year Computer Science and Engineering student proficient in **C++**, **Python**, and **SQL**. Developed a web-based decision support platform that achieved an **R2 ≈ 0.97** for demand forecasting. Eager to contribute to impactful engineering projects in a dynamic environment.

EDUCATIONS

Sardar Vallabhbhai National Institute of Technology (SVNIT) | Surat, Gujarat — Aug 2023 - May 2027
Bachelor of Technology in Computer Science and Engineering — *CGPA : 8.76/10.0*
- Expected graduation in **May 2027**

Netaji Model School | Vadodara, Gujarat — Jun 2022 - Apr 2023
Senior Secondary (GHSEB, Science) — *Percentage : 82%*
- Graduated in **April 2023** with **82%**

PROJECTS

Self-RAG - Self-Reflective Retrieval-Augmented Generation | Github — 2026 - 2026
Developer
- Built a RAG pipeline with adaptive retrieval and self-critique nodes (`[Retrieve]`, `[IsSup]`, `[IsUse]`) that verify answers are grounded before responding.
- Chunked PDFs into a FAISS vector store using HuggingFace embeddings, with LangGraph orchestrating retrieval, relevance, and support-verification stages.
- Benchmarked vs. baseline always-retrieve RAG on 15 questions: 100% accuracy vs. 0% on out-of-scope queries by correctly bypassing unneeded retrieval calls.
- Eliminated hallucinated responses via a grounding-verification refusal mechanism that rejects ungrounded answers, trading ~8x latency for reliability.

Technologies / Tools Used : *Python, Langgraph, FAISS, HuggingFace embeddings, PyPDF, Pydantic., lagnchain*

MedFL: Privacy-Preserving Federated Learning Platform | Github — 2026
Developer
- Built the aggregation service for a federated learning platform that trains a shared model across multiple hospital nodes without sharing raw patient data.
- Implemented ISH-weighted aggregation, using Earth Mover's Distance to measure data representativeness, combined with FedProx to reduce model drift from non-IID clinical data.
- Achieved **97.22% global model accuracy** after **5 federated rounds**, up from **75.69% in round 1**.
- Correctly upweighted the most statistically representative hospital in aggregation (**42% vs. ~29% for others**), improving on plain sample-count-based averaging (FedAvg).

Technologies / Tools Used : *Python, PyTorch, Docker, Federated Learning*

Retail Stockout Prediction & Inventory Transfer System | Github — 2025
Developer
- Built Sirn, a platform combining sales demand forecasting with automated inter-store inventory redistribution; Top 11 finalist at Walmart Sparkathon 2025.
- Trained and benchmarked Random Forest, XGBoost, and LightGBM on a 421,570-row Walmart sales dataset using a 17-feature pipeline, tested on 84,314 held-out rows.
- Achieved 97.52% R^2 with Random Forest and 95.88% R^2 with XGBoost; selected LightGBM for deployment (0.29MB vs. Random Forest's 2.92GB) at 90.89% R^2.
- Built a SARIMA forecasting model with 52-week walk-forward validation, achieving 89.86% forecast accuracy (10.14% MAPE).
- Engineered the inventory redistribution engine matching surplus and shortage stores via Haversine-distance transfer logic and a cost-effectiveness scoring formula.

Technologies / Tools Used : *React, TypeScript, Supabase, Python, Scikit-learn, web applications, APIs, unit testing*

SKILLS

Core CS fundamentals :	Data Structures, Algorithms, Object Oriented Programming, Computer Networks, Operating system, Design and Analysis of Algorithms
Databases :	MongoDB, PostgreSQL, MySQL, Redis
Frameworks & Libraries :	React.js, Node.js, Express.js, Next.js, HTML, CSS
Programming Languages :	C++, Python, JavaScript, TypeScript, SQL
Tools & Platforms :	Docker, GitHub, langchain, langgraph, Linux

AWARDS & ACHIEVEMENTS

- Selected among the top **11 teams** out of **7,000+ participants** in a national innovation hackathon.
- Attended **Amazon Machine Learning Summer School**.